

SOC Incident Response Playbook

Project 5 - SOC & SIEM Threat Hunting & Incident Response with Wazuh

Purpose

This playbook defines a repeatable SOC response process for security events observed in the Project 5 lab. It is designed to guide alert triage, investigation, containment planning, eradication, recovery, and lessons learned using Wazuh and endpoint evidence.

Scope

The playbook covers the security scenarios validated in the lab: repeated authentication failures, SSH authentication activity, privileged Windows logons, Linux sudo-to-root activity, Windows account creation or modification, local Administrators group changes, and controlled data-transfer indicators.

Core Response Workflow

Alert -> Triage -> Investigation -> Correlation -> Containment -> Eradication -> Recovery -> Lessons Learned

1. Alert Triage and Identification

- Review the Wazuh alert timestamp, rule ID, rule description, severity, agent name, and agent IP address.
- Identify the affected user account and endpoint.
- Determine whether the event involves authentication, privilege escalation, account manipulation, or data transfer.
- Prioritize repeated authentication failures and high-severity privilege/account-change alerts for immediate investigation.
- Preserve relevant event details before making changes to the affected system.

2. Investigation and Correlation

- Use Wazuh Threat Hunting to filter by rule ID, Windows Event ID, agent, user, and timestamp.
- Correlate related Windows Security events, Linux journal/audit events, authentication activity, and privilege-related events.
- Validate SIEM alerts against endpoint evidence where available.
- Build an event timeline to determine the sequence of activity.
- Classify the event as expected lab activity, suspicious activity requiring monitoring, or a security incident requiring response.

3. Scenario Response Procedures

Scenario	Indicators	Recommended Response
Repeated authentication failures / brute-force	Multiple failed logons, repeated SSH attempts, authentication-failure alerts	Identify source and target; verify whether attempts are authorized; restrict or block the source if malicious; review for a successful login after failures; reset credentials if compromise is suspected.
Privileged Windows logon	Event ID 4672 or related privilege alerts	Validate the account and reason for elevated access; review surrounding logon activity; terminate unauthorized sessions; disable or restrict the account if necessary.

Scenario	Indicators	Recommended Response
Linux sudo-to-root	Wazuh Rule 5402; sudo session events	Confirm the user and executed command; compare against authorized administrative activity; revoke unnecessary sudo access and investigate unauthorized elevation.
Account creation / modification	Windows Events 4720, 4722, 4738	Verify whether the account change was authorized; disable unauthorized accounts; review subsequent logons and privilege changes; preserve evidence.
Local Administrators group change	Event ID 4732; Wazuh Administrators Group Changed alert	Identify the added account and actor; remove unauthorized membership; investigate related account and authentication events; review administrative access.
Potential data transfer / exfiltration	Unexpected transfer tools, audit events, unusual outbound transfer activity	Validate the destination and data involved; restrict suspicious outbound communication; preserve logs; isolate the endpoint when warranted; investigate the initiating account/process.

4. Containment

- Isolate an affected endpoint when continued activity could cause additional impact.
- Disable or restrict suspicious accounts and terminate unauthorized sessions.
- Block or restrict confirmed malicious network sources or destinations.
- Preserve Wazuh and endpoint evidence before destructive remediation actions.
- In the controlled Project 5 lab, stop the test activity after detection and validation.

5. Eradication

- Remove unauthorized accounts and group memberships.
- Remove malicious or unauthorized processes and persistence mechanisms if identified.
- Correct insecure configurations and excessive privileges.
- Reset credentials when account compromise is suspected.
- Confirm that the original cause of the incident has been addressed.

6. Recovery

- Validate endpoint security and restore required services.
- Confirm legitimate account access and secure configurations.
- Reconnect isolated systems only after validation.
- Continue enhanced Wazuh monitoring for recurrence.
- Verify that normal operations have resumed without additional alerts related to the incident.

7. Evidence to Record

- Incident date and time
- Affected endpoint and IP address
- Affected user account
- Wazuh rule ID and description
- Windows Event ID or Linux log source
- Alert severity
- Relevant commands/processes
- Timeline of related events
- Containment and remediation actions
- Final disposition and lessons learned

8. Lessons Learned

After each incident or controlled validation, review whether logging coverage, Wazuh detection rules, alert severity, endpoint visibility, and response procedures should be improved. Authentication and privilege events should be correlated rather than reviewed independently, and high-severity account changes should receive prompt analyst attention.

9. Incident Record Template

Field	Record
Incident / Alert	
Date and Time	
Affected Endpoint	
User / Account	
Rule / Event ID	
Severity	
Summary of Activity	
Investigation Findings	
Containment Action	
Eradication Action	
Recovery Validation	
Final Status	
Lessons Learned	

Conclusion

This playbook converts the Project 5 detection and threat-hunting workflow into a structured incident-response process. It provides a practical reference for moving from a Wazuh alert through triage, investigation, containment, eradication, recovery, and post-incident improvement.